Dimensionality Reduction 1

Introduction to Data Mining

Seoul National University

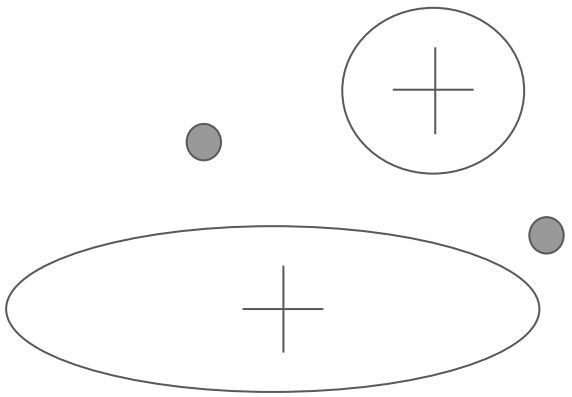
Spring 2026

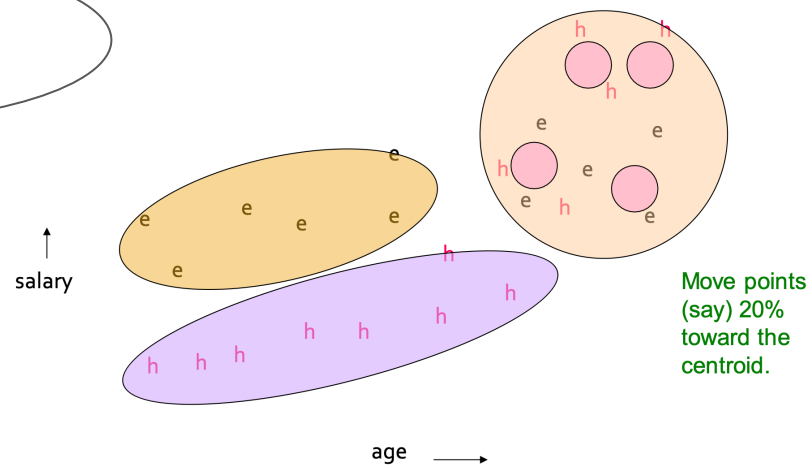
Announcements

- **Assignments:**
  - HW1 (deadline: April 9)
  - HW2 (deadline: April 30)
- **Midterm:**
  - **When:** April 23 at 9:30 – 10:45
  - **Where:** 302-208 and 302-209 (students will be separated)
  - **How:** Closed-book; no calculators allowed

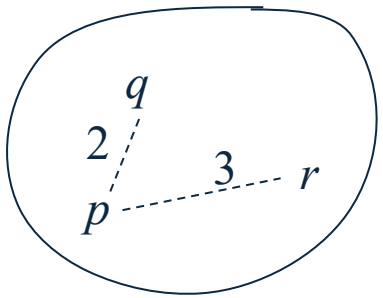
Recap

- BFR Algorithm
  - Cluster representation
  - Three classes of sets
- CURE Algorithm
  - Representative points
- GRGPF Algorithm
  - Rowsum
  - Estimation of a rowsum





$$rowsum \text{ of } p = \sum_{c \in C} d(p, c)^2$$



Outline

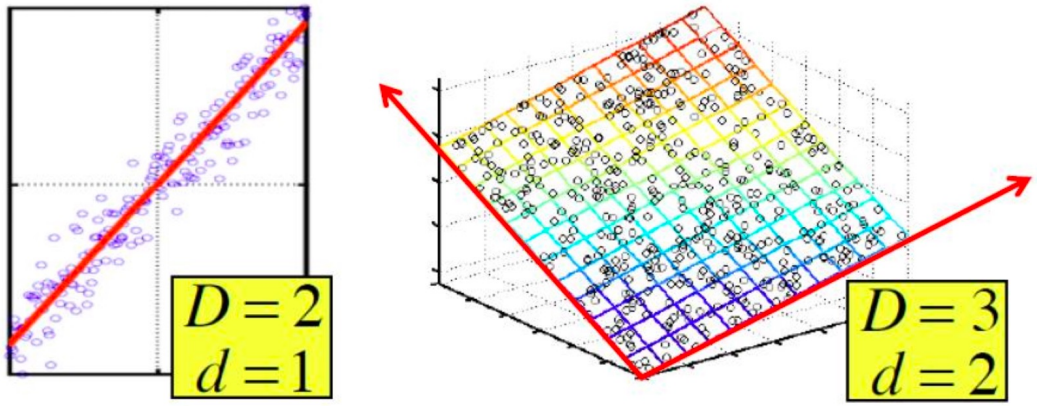
1. **<u>Dimensionality Reduction</u>**
2. Eigenvalues and Eigenpairs
3. Power Iteration
4. Principal Component Analysis

Data as a Large Matrix

- Many sources of data can be viewed as an $m \times n$ matrix.
  - m is the number of data points (or samples/instances).
  - n is the number of features (or attributes).
- Consider a health dataset as an example.
  - Each person is <height, weight, BMI, blood pressure, age, etc.>
  - Features are flexible; they can be newly created or discarded.

Data Manifold

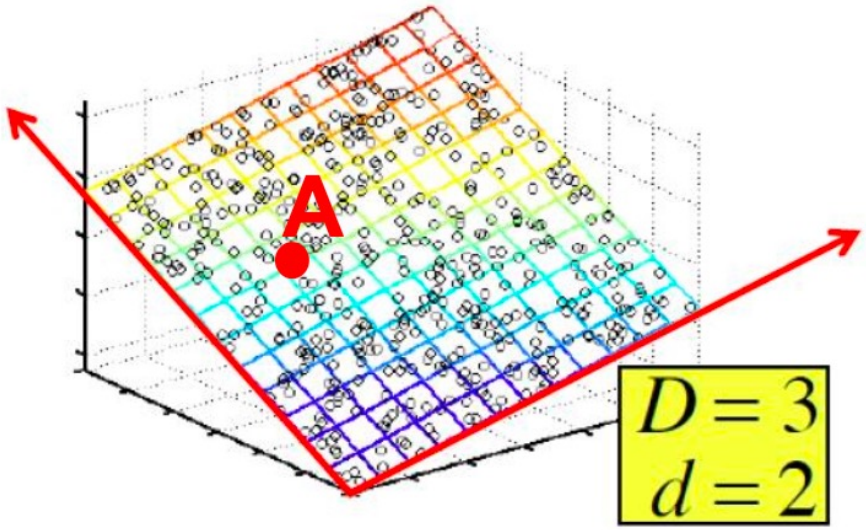
- The **manifold** of data is not necessarily an n-dimensional space.
  - The size of manifold = The number of **latent factors**.
  - E.g., representing a person with their weight, height, and BMI.
- We find these latent factors through **dimensionality reduction.**



Source: Stanford CS246 (2022)

Rank of a Matrix

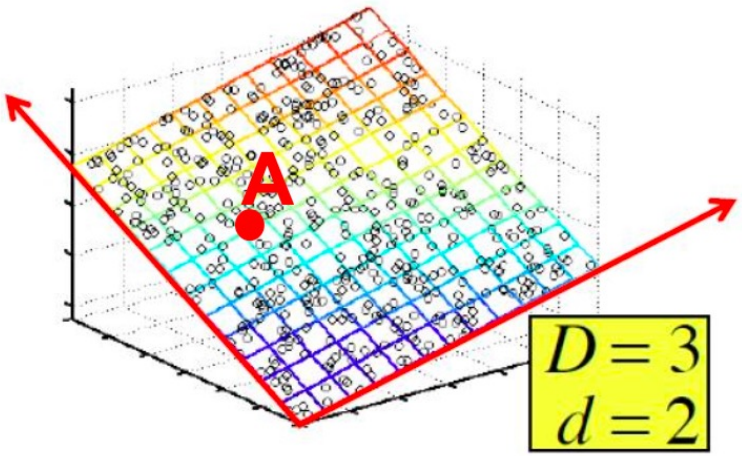
- The **rank** of a matrix M = The number of latent factors.
  - Rank is the # of linearly independent rows (or columns) of M.
  - **Independence:** There is no nonzero linear sum of rows that is $\mathbf{0}$.
- Consider 3 points in a 3D space:
  - $\begin{bmatrix} A \\ B \\ C \end{bmatrix} = \begin{bmatrix} 1 & 2 & 1 \\ -2 & -3 & 1 \\ 3 & 5 & 0 \end{bmatrix}$ (1 row per point)
- Rank is two, since
  - $A - B - C = \mathbf{0}$ (in terms of rows)
  - $5c_1 - 3c_2 + c_3 = \mathbf{0}$ (in terms of columns)



Source: Stanford CS246 (2022)

Rank of a Matrix

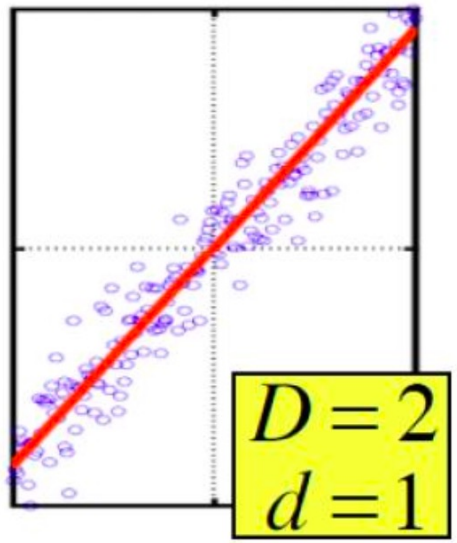
- Then, we can rewrite the coordinates efficiently.
- Let's consider $A = [1 \;\; 2 \;\; 1]$.
- Old basis vectors: $[1\ 0\ 0], [0\ 1\ 0], [0\ 0\ 1]$.
  - $A = 1u_1 + 2u_2 + 1u_3$.
- **New basis vectors:** $[1\ 2\ 1], [-2 \;\; -3\ 1]$.
  - $A = 1u_1 + 0u_2$.
  - M has new coordinates: A: $[1\ 0]$, B: $[0\ 1]$, C: $[1 \;\; -1]$.
  - We reduced the number of dimensions!



Source: Stanford CS246 (2022)

Dimensionality Reduction

- **Goal:** Discover **the actual axes** of data.
  - Follow the red line (i.e., the manifold).
  - Represent each point with 1 coordinate.
- By doing this, we incur a **bit of error**.
  - Since the points do not exactly lie on the line.
  - There will be no error if the rank is one.



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Outline

1. Dimensionality Reduction
2. **<u>Eigenvalues and Eigenpairs</u>**
3. Power Iteration
4. Principal Component Analysis

Terms on Orthogonality

- A vector v is a **unit vector** if $\|v\| \equiv \|v\|_2 \equiv \mathrm{sqrt}\left(\sum_i v_i^2\right) = 1$.
- Two vectors v_1 and v_2 are **orthogonal** if $v_1^\top v_2 = 0$.
- Two vectors v_1 and v_2 are **orthonormal** if orthogonal and unit.
- A matrix A is **orthogonal** if its rows/columns are orthonormal.
  - Then, $AA^\top = A^\top A = I$ and $A^\top = A^{-1}$.

Eigenvalues and Eigenvectors

- Let M be an $n \times n$ square matrix, λ be a scalar, and e be a vector.
- λ is an **eigenvalue** of M and e is the corresponding **eigenvector** if:

$$Me = \lambda e$$

- A pair of λ and e is called an **eigenpair**.
- The eigenvector with the largest $|\lambda|$ is the **principal eigenvector**.

Ambiguity of Eigenvectors

- If e is an eigenvector, ce is also an eigenvector (c: constant).
  - Why? $Mce = cMe = c\lambda e = \lambda ce.$
  - Thus, e and ce correspond to the same eigenvalue.
- To avoid **ambiguity**, we require
  - Every eigenvector e to be a unit vector, i.e., $\|e\| = 1$.
  - The first nonzero component of an eigenvector to be positive.

$$\underset{M}{\begin{bmatrix} 3 & 2 \\ 2 & 6 \end{bmatrix}} \underset{e}{\begin{bmatrix} 1/\sqrt{5} \\ 2/\sqrt{5} \end{bmatrix}} \underset{\lambda}{= 7} \underset{e}{\begin{bmatrix} 1/\sqrt{5} \\ 2/\sqrt{5} \end{bmatrix}}$$

Positive

Unit vector

Eigendecomposition

- **Eigendecomposition** of a square matrix M is defined as

$$M = Q\Lambda Q^{-1}.$$

  - Q is the $n \times n$ matrix whose i-th column is the i-th eigenvector of M.
  - Λ is the diagonal matrix whose diagonal elements are the eigenvalues.
- **Note:** Eigendecomposition is not for all matrices.

Diagonalizable Matrix

- A square matrix D is **diagonal** if $i \neq j \Longrightarrow d(i,j) = 0$.
- A square matrix M is **diagonalizable** if
  - **Informal:** it is an *almost diagonal* matrix.
  - **Formal:** there exists an invertible matrix P and a diagonal matrix D s.t.

$$P^{-1}MP = D.$$

Independent Eigenvectors

- Any diagonalizable matrix M has eigendecomposition.
- Why? M always has n linearly independent eigenvectors.
  - $P^{-1}MP = D$ for some invertible P and diagonal D.
  - Then, $MP = PD \Longrightarrow Mv_i = P(\lambda_i e_i) = \lambda_i v_i$.
    - v_i is the i-th column of P.
    - λ_i is the i-th diagonal entry of D.
    - e_i is the i-th column of the identity matrix I.
  - Therefore, the columns of P are eigenvectors of M.
  - Since P is invertible, they are linearly independent.

Eigenvectors of a Symmetric Matrix

- Let S be an $n \times n$ **real symmetric matrix**, i.e., $S = S^\top$.
  - No complex matrices during our course.
- **Property 1:** S has n orthonormal eigenvectors.
  - Symmetry lets S act independently along orthogonal directions.
- **Property 2:** S is diagonalizable.
  - n linearly independent eigenvectors $\Longrightarrow$ diagonalizable.
- **Property 3:** Q in $S = Q\Lambda Q^{-1}$ is orthonormal.
  - Due to P1 & P2.

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1. Dimensionality Reduction
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How to Perform Eigendecomposition

- **Q:** How can we actually perform eigendecomposition?
- Two popular approaches: **QR algorithm** and **power iteration**.
  - QR algorithm is a standard way but slow for large matrices.
  - Power iteration is fast but finds only the largest eigenvalues.
- We'll focus on power iteration as we assume a large matrix.

Power Iteration

- **Power iteration** finds the principal eigenvector of M efficiently.
  - Again, M is an $n \times n$ diagonalizable matrix.
- **How it works:**
  - **Step 1:** Start with any nonzero vector x_0 of length n.
  - **Step 2:** Construct $x_{k+1} = \frac{Mx_k}{\|Mx_k\|}$ for $k = 0, 1, \cdots$.
  - **Step 3:** Stop when consecutive x_k's show little change.

Power Iteration: Example

$$M = \begin{pmatrix} 1 & 2 \\ 2 & 3 \end{pmatrix} \quad \mathbf{x}_0 = \begin{matrix} 1 \\ 1 \end{matrix}$$

$$\frac{M\mathbf{x}_0}{||M\mathbf{x}_0||} = \begin{pmatrix} 3 \\ 5 \end{pmatrix} / \sqrt{34} = \begin{pmatrix} 0.51 \\ 0.86 \end{pmatrix} = \mathbf{x}_1$$

$$\frac{M\mathbf{x}_1}{||M\mathbf{x}_1||} = \begin{pmatrix} 2.23 \\ 3.60 \end{pmatrix} / \sqrt{17.93} = \begin{pmatrix} 0.53 \\ 0.85 \end{pmatrix} = \mathbf{x}_2$$

.....

Finding an Eigenvector

- The converged x is approximately an eigenvector of M.
  - **Why?** $Mx \approx x$.
- The corresponding eigenvalue is $\lambda = x^\top Mx$.
  - **Why?** $x^\top Mx = x^\top \lambda x = \lambda x^\top x = \lambda \|x\|^2 = \lambda$ (x is a unit vector).
- **Example:** If we take $x = (0.53, 0.85)$, then

$$\lambda = [0.53\ 0.85] \begin{bmatrix} 1 & 2 \\ 2 & 3 \end{bmatrix} \begin{bmatrix} 0.53 \\ 0.85 \end{bmatrix} = 4.25$$

Finding the Principal Eigenvector

- The converged x is approximately the **principal eigenvector** of M.
  - Suppose that the eigenvalues are $|\lambda_1| > |\lambda_2| \geq \cdots \geq |\lambda_n|$.
  - Let the corresponding eigenvectors be $v_1, v_2, \cdots, v_n$.
  - The initial vector can be represented as $x_0 = c_1 v_1 + c_2 v_2 + \cdots + c_n v_n$.
  - Power iteration works as $x_k = c_1 \lambda_1^k v_1 + c_2 \lambda_2^k v_2 + \cdots + c_n \lambda_n^k v_n$.
  - By factoring out λ_1^k, we have

$$x_k = \lambda_1^k \left(c_1 v_1 + c_2 \left(\frac{\lambda_2}{\lambda_1}\right)^k v_2 + \cdots + c_n \left(\frac{\lambda_n}{\lambda_1}\right)^k v_n \right).$$

  - As $k \to \infty$, all $(\lambda_i/\lambda_1)^k$ for $i > 1$ goes to 0.

Finding More Eigenpairs

- Create a new matrix $M^* = M - \lambda x x^\top$.
  - Eliminate the influence of x by setting its associated eigenvalue to 0.
- Recursively use power iteration on M^*, and so on.
- **Example:**

$$M^* = \begin{bmatrix} 1 & 2 \\ 2 & 3 \end{bmatrix} - 4.25 \begin{bmatrix} 0.53 \\ 0.85 \end{bmatrix} [0.53\ 0.85] = \begin{bmatrix} -0.19 & 0.09 \\ 0.09 & 0.07 \end{bmatrix}$$

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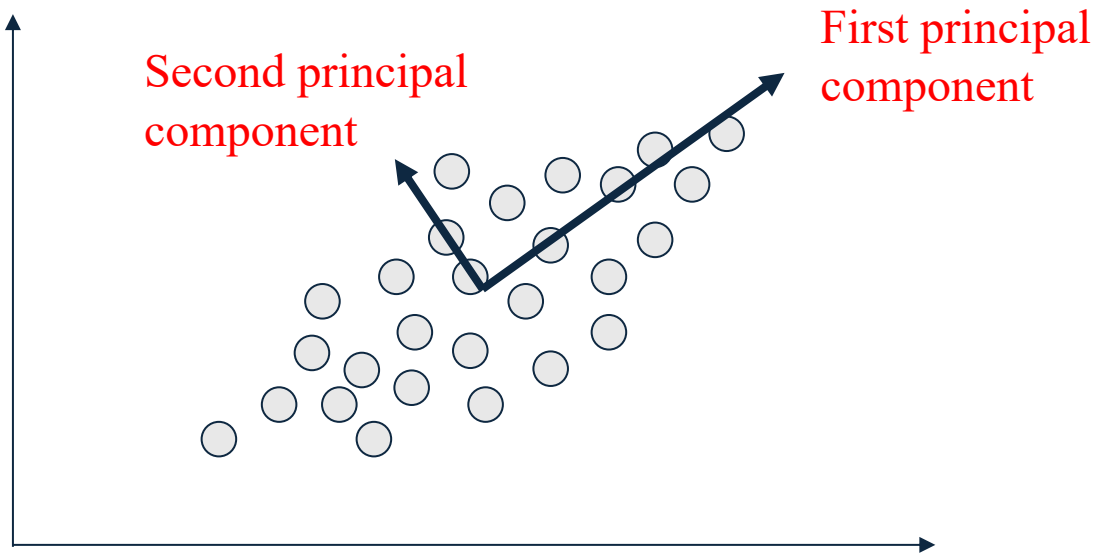
1. Dimensionality Reduction
2. Eigenvalues and Eigenpairs
3. Power Iteration
4. **<u>Principal Component Analysis</u>**

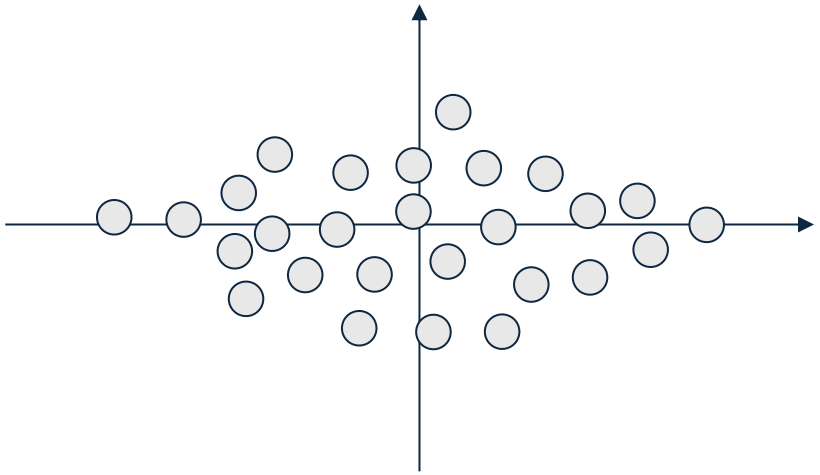
Principal Component Analysis

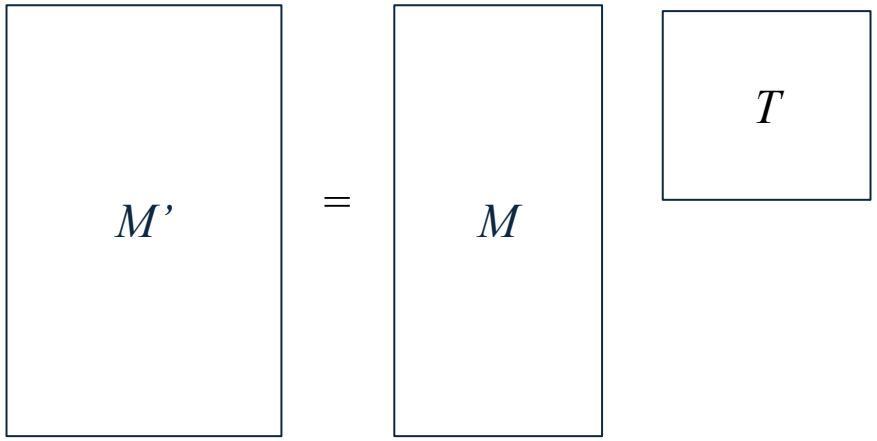
- **Principal component analysis (PCA):**
  - Given n-dimensional data, it finds the new n axes such that:
  - **1st dimension** shows the greatest variance.
  - **2nd dimension** shows the next-greatest variance, orthogonal to the 1st.



Output of PCA

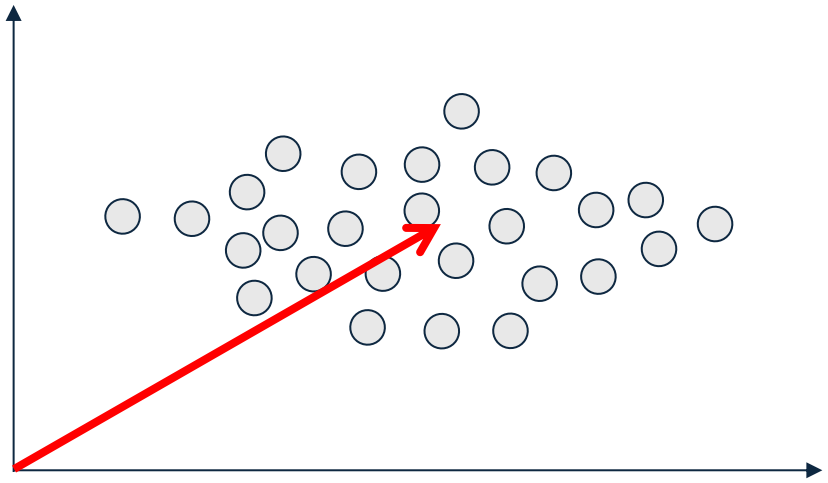
- **Given** an $m \times n$ data matrix M with **column-wise zero mean**.
- **Create** an $n \times n$ **transformation matrix** T to make $M' = MT$.
  - The i-th column of T creates the i-th PC.
  - Dimensionality reduction is done if we use the first r columns.





Why Zero Mean?

- What will happen if the input M is not zero-centered?
- The first PC is likely to point the mean of data.
  - Since it introduces a large (artificial) variance.
- Mean-centering (or feature standardization) is essential.



Optimization Problem

- **Q:** How can we efficiently find the first column t_1 of T?
- Let's define the variance of data for a column t as

$$\mathrm{var}(t) \equiv \sum_i (M_i t - 0)^2 = \|Mt\|^2 = (Mt)^\top Mt = t^\top M^\top Mt.$$

  - M_i is the i-th row of M.
- Then, to maximize variance, $t_1 = \mathrm{argmax}_{\|t\|=1} \mathrm{var}(t)$.

Rayleigh Quotient

- **Observation 1:** $\mathrm{var}(t) = t^\top M^\top M t$ is between $\lambda_{\min}$ and $\lambda_{\max}$.
  - $\lambda_{\min}$ is the smallest eigenvalue of $M^\top M$.
  - $\lambda_{\max}$ is the largest eigenvalue of $M^\top M$.
- **Proof:**
  - $t^\top M^\top M t = t^\top (Q \Lambda \mathrm{Q}^\top) t = (t^\top Q) \Lambda (Q^\top t)$.
    - $M^\top M$ is a symmetric matrix and thus is diagonalizable.
  - Let $y \equiv Q^\top t$; then, y is a unit vector.
    - Why? $\|y\|^2 = y^\top y = (Q^\top t)^\top (Q^\top t) = t^\top Q Q^\top t = t^\top t = 1$.
  - Thus, $y^\top \Lambda \mathrm{y} = \sum_i \lambda_i y_i^2$ is a weighted average of λ_i's.

Rayleigh Quotient

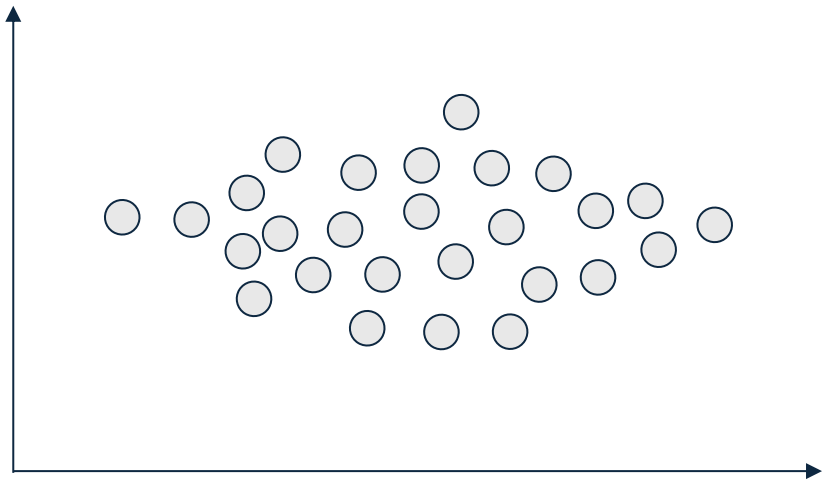
- **Observation 2:** $\mathrm{var}(t) = t^\top M^\top M t = \lambda_{\max}$ when $t = v_1$.
  - v_1 is the principal eigenvector of $M^\top M$.
- **Proof:**
  - $\mathrm{var}(t) = \sum_i \lambda_i y_i^2$ is a weighted average of eigenvalues λ_i's.
  - The maximum is achieved when $|y_1| = 1$ and $|y_{i>1}| = 0$.
  - If $|y_1| = 1$, then $|y_1| = |(Q^\top t)_1| = \left|v_1^\top t\right| = 1$.
  - Since v_1 and t are both unit vectors, we conclude that $t = v_1$.

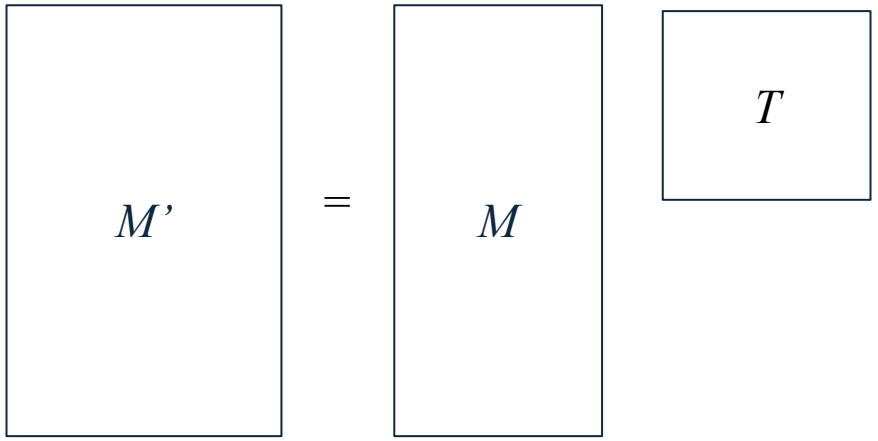
Optimization Problem

- **Q:** How can we efficiently find the first column t_1 of T?
- **Answer:** t_1 is the principal eigenvector of $M^\top M$ with $\lambda_{\max}$.
  - Based on Observations 1 and 2.
- We can run power iteration to find t_1, and t_2, and so on.
  - t_2 is the second principal eigenvector of $M^\top M$.
  - $M^\top M$ is often called the *covariance matrix* of (centered) data.

Dimensionality Reduction by PCA

- PCA is widely used for dimensionality reduction.
- By keeping the first r columns in M', not all n columns.
  - The chosen columns have the greatest variances.
  - Thus, M' can represent the "information" even with $r < n$.





Summary

1. Dimensionality Reduction
   - Latent factors
2. Eigenvalues and eigenpairs
   - Eigendecomposition
3. Power Iteration
   - Principal eigenvectors
4. Principal Component Analysis
   - Principal components
   - Transformation matrix